

1. RSA ALGORITHM WITH NUMERICAL EXAMPLE

DES OVERVIEW

DES is a symmetric-key algorithm. It is a Feistel network. It takes a 64-bit plaintext and a 64-bit key and produces a 64-bit ciphertext. It is a block cipher. It is a stream cipher. It is a public-key algorithm. It is a private-key algorithm.

DES ALGORITHM

1. Plaintext (64 bits) is split into two 32-bit halves, L and R.

2. L and R are processed by a function f, which uses a 48-bit round function key.

3. The result is XORed with the other half, and the halves are swapped.

4. This process is repeated for 16 rounds.

5. The final result is the ciphertext (64 bits).

NUMERICAL EXAMPLE

Let's take a simple example. Plaintext = 10101010101010101010101010101010. Key = 10101010101010101010101010101010.

Step 1: Initial permutation (IP). Plaintext is split into two 32-bit halves, L and R.

Step 2: Round function f. L and R are processed by a function f, which uses a 48-bit round function key.

Step 3: XOR and swap. The result is XORed with the other half, and the halves are swapped.

Step 4: Repeat for 16 rounds.

Step 5: Final permutation (FP). The final result is the ciphertext (64 bits).

2. AES STRUCTURE AND TRANSFORMATION STEPS

AES OVERVIEW

AES is a symmetric-key algorithm. It is a Feistel network. It takes a 128-bit plaintext and a 128-bit key and produces a 128-bit ciphertext. It is a block cipher. It is a stream cipher. It is a public-key algorithm. It is a private-key algorithm.

AES STRUCTURE

1. Plaintext (128 bits) is split into two 64-bit halves, L and R.

2. L and R are processed by a function f, which uses a 48-bit round function key.

3. The result is XORed with the other half, and the halves are swapped.

4. This process is repeated for 10 rounds.

5. The final result is the ciphertext (128 bits).

AES TRANSFORMATION STEPS

1. Substitution: Each byte of the data is replaced by another byte using a substitution table.

2. Shift Rows: Each row of the data is shifted by a certain number of positions.

3. Mix Columns: Each column of the data is multiplied by a certain matrix.

4. Add Round Key: The result is XORed with a round key.

3. SQL INJECTION ATTACK WITH PREVENTION

SQL INJECTION ATTACK

SQL injection is a type of attack where an attacker injects malicious SQL code into a web application. This can allow the attacker to access sensitive data, modify data, or even delete data.

PREVENTION

1. Use parameterized queries (prepared statements).

2. Use stored procedures.

3. Use input validation.

4. Use least privilege.

5. Use web application firewalls (WAFs).

6. Use security audits.

7. Use security training.

8. Use security patches.

9. Use security monitoring.

10. Use security logging.

1. TYPES OF SECURITY ATTACKS

ACTIVE ATTACKS

Active attacks involve the attacker modifying or destroying data. Examples include: Denial of Service (DoS), Man-in-the-Middle (MitM), and Spoofing.

PASSIVE ATTACKS

Passive attacks involve the attacker monitoring or intercepting data. Examples include: Eavesdropping and Traffic Analysis.

2. SECURITY SERVICES AND MECHANISMS

SECURITY SERVICES

1. Confidentiality: Ensuring that information is not disclosed to unauthorized individuals.

2. Integrity: Ensuring that information is not altered or destroyed.

3. Availability: Ensuring that information is accessible to authorized individuals.

4. Authentication: Ensuring that the identity of the user or system is verified.

5. Non-repudiation: Ensuring that the user or system cannot deny the action.

SECURITY MECHANISMS

1. Access Control: Controlling access to resources.

2. Encryption: Protecting data by converting it into a form that is unreadable.

3. Authentication: Verifying the identity of the user or system.

4. Authorization: Granting or denying access to resources.

5. Auditing: Monitoring and logging system activity.

3. CIA TRIAD

The CIA triad is a model for information security. It consists of three pillars: Confidentiality, Integrity, and Availability.

CONFIDENTIALITY

Ensuring that information is not disclosed to unauthorized individuals.

INTEGRITY

Ensuring that information is not altered or destroyed.

AVAILABILITY

Ensuring that information is accessible to authorized individuals.

1. DIFFERENCE BETWEEN ACTIVE AND PASSIVE ATTACKS

ACTIVE ATTACKS

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PASSIVE ATTACKS

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2. TYPES OF MALICIOUS SOFTWARE

(a) VIRUS

A virus is a malicious program that can replicate itself and spread to other files.

(b) WORM

A worm is a malicious program that can replicate itself and spread to other systems.

(c) TROJAN HORSE

A Trojan horse is a malicious program that appears to be legitimate but actually contains malicious code.

3. BOTNET AND SPYWARE

(a) BOTNET

A botnet is a group of infected computers (bots) controlled by an attacker (botmaster).

(b) SPYWARE

Spyware is malicious software that secretly monitors user activities and sends the information to the attacker.

1. DES (DATA ENCRYPTION STANDARD)

DES OVERVIEW

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DES ALGORITHM - GENERAL DESIGN

1. Plaintext (64 bits) is split into two 32-bit halves, L and R.

2. L and R are processed by a function f, which uses a 48-bit round function key.

3. The result is XORed with the other half, and the halves are swapped.

4. This process is repeated for 16 rounds.

5. The final result is the ciphertext (64 bits).

2. COMPARISON: SYMMETRIC vs ASYMMETRIC CRYPTOGRAPHY

SYMMETRIC CRYPTOGRAPHY

Uses the same key for encryption and decryption. Examples include: DES, AES, and Blowfish.

ASYMMETRIC CRYPTOGRAPHY

Uses a pair of keys: a public key for encryption and a private key for decryption. Examples include: RSA, ECC, and ElGamal.

1. DEFINITION OF CRYPTOANALYSIS

Cryptoanalysis is the study and analysis of cryptographic systems and protocols to discover weaknesses and break them to obtain the original (plaintext) information without knowing the secret key.

2. STRENGTH OF DES (DATA ENCRYPTION STANDARD)

OVERVIEW

DES is a symmetric-key block cipher that uses a 56-bit key to encrypt 64-bit data blocks. It was widely used and proved to be strong at the time of its design.

STRENGTHS

- Well-tested and widely analyzed.
- Good avalanche effect.
- Confusion and diffusion.
- 56-bit key was strong enough against brute force when introduced (1977).
- Simple and efficient design.
- Used in many secure applications (ATM, banking, etc.).

LIMITATION

- 56-bit key is now too small. Brute force attack is possible (2⁵⁶ possibilities).
- DES is replaced by 3DES and AES.

DES STRUCTURE (Feistel Network)

1. Plaintext (64 bits) is split into two 32-bit halves, L and R.

2. L and R are processed by a function f, which uses a 48-bit round function key.

3. The result is XORed with the other half, and the halves are swapped.

4. This process is repeated for 16 rounds.

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3. DIFFERENCE: HASH vs ENCRYPTION

HASH FUNCTION

Converts data of any size to a fixed-size hash value. It is irreversible.

ENCRYPTION

Converts plaintext into ciphertext using a key. It is reversible.

4. APPLICATIONS OF HASH FUNCTION

1. Data Integrity: Verify that data has not been altered during transmission or storage.

2. Digital Signatures: Verify the authenticity and integrity of messages.

3. Data Deduplication: Identify duplicate files by comparing hash values instead of entire files.

4. Password Storage: Store passwords as hash values, not in plain text.

5. Message Authentication (MAC): Used in HMAC to verify authenticity and integrity of messages.

6. Blockchain Technology: Hashes are used to link blocks and ensure data integrity in blockchain.

7. File/Software Verification: Verify the authenticity of downloaded files or software using checksum (hash).

1. BUFFER OVERFLOW ATTACK

DEFINITION
A buffer overflow occurs when a program writes more data to a buffer than it was designed to hold, causing the program to overwrite adjacent memory locations and potentially crash or execute arbitrary code.

HOW BUFFER OVERFLOW OCCURS
A buffer overflow occurs when a program writes more data to a buffer than it was designed to hold, causing the program to overwrite adjacent memory locations and potentially crash or execute arbitrary code.

2. SPOOFING AND SNIFFING ATTACKS

A. SPOOFING ATTACK
Spoofing is an attack where an attacker pretends to be a legitimate user or system to gain unauthorized access to a system or network.

B. SNIFFING ATTACK
Sniffing is an attack where an attacker intercepts and steals sensitive information, such as passwords, credit card numbers, or other confidential data, as it travels across a network.

3. SPLITTING AND JOINING ATTACKS

A. SPLITTING ATTACK
A splitting attack is a type of cryptographic attack where an attacker divides a ciphertext into smaller parts and analyzes them individually to find the plaintext.

B. JOINING ATTACK
A joining attack is a type of cryptographic attack where an attacker combines multiple ciphertexts to find the plaintext.

1. DES (DATA ENCRYPTION STANDARD)

OVERVIEW
DES is a symmetric-key algorithm for encrypting and decrypting data. It uses a 56-bit key and operates on 64-bit blocks of data.

DES ALGORITHM - OVERALL DESIGN
The DES algorithm consists of an initial permutation, 16 rounds of function application, and a final permutation. Each round function is a complex operation involving substitution and permutation.

DES ROUND PROCESS SUMMARY
The DES round process involves a function application (F-function) and a permutation (P-function). The F-function is a complex operation involving substitution and permutation.

2. COMPARISON: SYMMETRIC VS ASYMMETRIC CRYPTOGRAPHY

Feature	Symmetric Cryptography	Asymmetric Cryptography
Key Management	Simple, but requires a secure channel for key distribution.	Complex, but allows for secure key distribution over an insecure channel.
Performance	Fast, but less secure.	Slower, but more secure.
Use Cases	Used for bulk data encryption and decryption.	Used for secure communication, digital signatures, and key exchange.

1. PHISHING

DEFINITION
Phishing is a social engineering attack where an attacker tricks a user into revealing sensitive information, such as passwords, credit card numbers, or other confidential data.

HOW PHISHING WORKS
An attacker sends a fraudulent email or message to a user, claiming to be a legitimate entity. The user is then directed to a fake website where they are asked to enter their sensitive information.

2. SESSION HIJACKING

DEFINITION
Session hijacking is an attack where an attacker steals a user's session ID and uses it to impersonate the user and access their account.

HOW SESSION HIJACKING WORKS
An attacker intercepts a user's session ID and uses it to impersonate the user and access their account.

3. COOKIES

DEFINITION
Cookies are small text files stored on a user's device by a web browser. They are used to store information about the user and their preferences.

HOW COOKIES WORK
A web server sends a cookie to a user's browser. The browser stores the cookie and sends it back to the server on subsequent requests.

1. DIGITAL SIGNATURE

DEFINITION
A digital signature is a cryptographic technique that provides a way to verify the authenticity and integrity of a message or document.

DIGITAL SIGNATURE PROCESS
The digital signature process involves creating a hash of the message and then signing the hash with a private key.

2. MAC (MESSAGE AUTHENTICATION CODE)

DEFINITION
A MAC is a short, fixed-length code generated using a secret key and a message. It is used to verify the integrity and authenticity of the message.

MAC GENERATION PROCESS
The MAC generation process involves applying a MAC function to a message and a secret key to produce a MAC value.

3. DOS ATTACK (DENIAL OF SERVICE)

DEFINITION
A Denial of Service (DoS) attack is an attempt to make a system or network unavailable to its intended users by overwhelming it with a flood of traffic.

HOW DOS ATTACKS WORK
An attacker sends a large volume of traffic to a system, causing it to become overloaded and unable to respond to legitimate requests.

1. CROSS-SITE SCRIPTING (XSS)

DEFINITION
Cross-Site Scripting (XSS) is a security vulnerability that allows an attacker to inject malicious scripts into a web page viewed by other users.

HOW XSS ATTACKS WORK
An attacker injects a malicious script into a web page. When a user visits the page, the script is executed, potentially stealing the user's session ID or other sensitive information.

2. FIREWALLS AND THEIR TYPES

DEFINITION
A firewall is a network security device that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

TYPES OF FIREWALLS
There are several types of firewalls, including packet-filtering firewalls, stateful inspection firewalls, and proxy firewalls.

3. INTRUSION DETECTION SYSTEM (IDS)

DEFINITION
An Intrusion Detection System (IDS) is a security system that monitors network traffic for signs of malicious activity and alerts administrators when a threat is detected.

HOW IDS WORKS
An IDS monitors network traffic and compares it to a database of known attack signatures. If a match is found, the IDS generates an alert.

1. RACE CONDITION

DEFINITION
A race condition is a type of synchronization error that occurs when the execution of a program depends on the relative timing of events that cannot be easily controlled.

HOW RACE CONDITIONS OCCUR
A race condition occurs when two or more processes access a shared resource and at least one of them modifies it. The final state of the resource depends on the order in which the processes execute.

2. ACCESS CONTROL PROBLEMS

DEFINITION
Access control problems occur when a system's access control mechanisms are flawed, allowing unauthorized users to access sensitive data or perform actions.

COMMON ACCESS CONTROL PROBLEMS
Common access control problems include weak authentication, poor authorization, and inadequate logging.

3. DIFFERENCE BETWEEN SSL AND TLS

DEFINITION
SSL (Secure Sockets Layer) and TLS (Transport Layer Security) are cryptographic protocols that provide secure communication over a network.

DIFFERENCES
SSL is an older protocol, while TLS is its successor. TLS is more secure and has better support for modern applications.

4. DNS SECURITY

DEFINITION
DNS (Domain Name System) security is the process of protecting the DNS infrastructure from attacks and ensuring the integrity and availability of DNS services.

HOW DNS SECURITY WORKS
DNS security is implemented through various measures, including DNSSEC (DNS Security Extensions) and secure DNS protocols.

1. What is Phishing?

Phishing is a social engineering attack in which an attacker tricks users into revealing sensitive information like usernames, passwords or credit card details by pretending to be a trustworthy entity.

2. Define SQL Injection.

SQL Injection is an attack technique in which attacker inserts malicious SQL queries into input fields to manipulate the database.

3. What is Cross Site Scripting (XSS)?

XSS is an attack in which attacker injects malicious scripts into webpages viewed by other fields. The script runs in the user's browser and can steal cookies, session tokens or other sensitive data.

4. What is Session Hijacking?

Session Hijacking is an attack where an attacker takes over a valid user session by stealing the session ID (cookie) and uses it to access the system as that user.

5. Define Firewall.

A Firewall is a network security system that monitors and controls incoming and outgoing network traffic based on predefined security rules.

6. Types of Firewalls.

- Packet Filtering Firewall:** Filters packets based on source, destination IP address, ports and protocols.
- Stateful Inspection Firewall:** Monitors the state of active connections and makes decisions.
- Application Level Gateway:** Works at application layer and acts as a proxy between client and server.
- Next Generation Firewall (NGFW):** Combines traditional firewall with additional features like IDS, IPS, application awareness.

7. What is Intrusion Detection System (IDS)?

An IDS is a security system that monitors network or system activities for malicious activities or policy violations and alerts the administrator.

8. Difference between Stateless and Stateful Firewall.

Feature	Stateless Firewall	Stateful Firewall
Meaning	Does not maintain any information about previous connections.	Maintains state information of active connections.
How it works	Decides based on packet header (IP, Port, Protocol).	Decides based on packet header + connection state.
Security	Less secure.	More secure.
Performance	Faster.	Slightly slower than stateless.
Example	Packet Filtering Firewall.	Stateful Inspection Firewall.

9. What is SSL?

SSL (Secure Sockets Layer) is a cryptographic protocol that provides secure communication over a network by ensuring confidentiality, integrity and authenticity.

10. Difference between SSL and TLS.

Feature	SSL	TLS
Full Form	Secure Sockets Layer	Transport Layer Security
Developed by	Netscape	IETF
Security	Less secure (older versions)	More secure
Current Use	Deprecated	Widely used (SSL is replaced by TLS)
Latest Version	SSL 3.0	TLS 1.3

11. What is HTTPS?

HTTPS (Hypertext Transfer Protocol Secure) is the secure version of HTTP. It uses TLS/SSL protocol to encrypt data between client and server.

12. Quick Summary

Phishing: Trick users to steal sensitive information.
SQL Injection: Inserting malicious SQL queries to access database.
XSS: Injecting malicious scripts in webpages.
Session Hijacking: Stealing session ID to take over session.
Firewall: Monitors and controls network traffic.
Types of Firewall: Packet Filtering, Stateful, Application Level Gateway, NGFW.
IDS: Monitors system/network and alerts on attacks.
SSL: Protocol for secure communication.
SSL vs TLS: TLS is more secure and latest.
HTTPS: Secure HTTP using SSL/TLS.
Stateless vs Stateful Firewall: Stateless - no state info. Stateful - maintains connection state.

1. Define IPsec.

IPsec (Internet Protocol Security) is a suite of protocols and algorithms for securing network communications by authenticating and encrypting each IP packet of a communication session.

2. What is SSH?

SSH (Secure Shell) is a cryptographic network protocol for operating network services over an unsecured network.

3. What is DNS Security?

DNS Security refers to protecting the Domain Name System (DNS) from attacks that could compromise the availability, confidentiality and integrity of DNS services.

4. What is a DoS attack?

A DoS (Denial of Service) attack is an attempt to make a computer or network resource unavailable to its intended users by overwhelming it with malicious traffic.

5. Define Buffer Overflow.

Buffer Overflow is a vulnerability that occurs when a program writes more data to a buffer than it can hold, causing the program to crash or execute arbitrary code.

6. What is Spoofing?

Spoofing is an attack in which an attacker impersonates as a legitimate entity by falsifying data to gain unauthorized access or steal information.

7. Define Sniffing attack.

Sniffing is a passive attack where an attacker intercepts and monitors network traffic to steal sensitive information.

8. What are Race Conditions?

Race Condition occurs when two or more processes simultaneously access and manipulate shared data, leading to unpredictable behavior.

9. Difference between Hash and Encryption.

Feature	Hash	Encryption
Reversible	No	Yes
Key size	Fixed size (Hash value)	Variable size (Plaintext)
Output	One-way (One-to-one)	Two-way (One-to-one)
Process	No key required	Key required
Example	SHA-256, MD5	AES, DES, RSA

10. Define Digital Signature.

Digital Signature is a mathematical technique used to verify the authenticity and integrity of a digital message or document.

11. What is Authentication?

Authentication is the process of verifying the identity of a user or system before granting access to resources.

1. Define Security Attack with example.

A Security Attack is any action that attempts to compromise the confidentiality, integrity or availability of information or resources.

Example: An attacker steals passwords by phishing email.

2. What are Security Services?

- Authentication
- Access Control
- Confidentiality
- Integrity
- Non-repudiation
- Availability

3. Explain CIA triad (Confidentiality, Integrity, Availability).

Confidentiality: Ensures information is accessible only to authorized users.

Integrity: Ensures information is accurate, complete and unaltered.

Availability: Ensures information and services are available when needed.

4. Difference between active and passive attacks.

Feature	Active Attack	Passive Attack
Goal	Affect, modify or destroy data or resources.	Monitor or observe data without changing it.
Example	DoS attack, Malware, Data modification.	Sniffing, Traffic analysis.
Detectability	Easily detected.	Difficult to detect.
Affects System	Yes	No

5. What is Malware? List types.

Malware (Malicious Software) is any software designed to harm, exploit or gain unauthorized access to a system.

Types: Virus, Worm, Trojan Horse, Ransomware, Spyware, Adware, Rootkit.

6. Difference between Virus and Worm.

Feature	Virus	Worm
Definition	Attaches itself to a host file/program.	Independent program.
Spread	Needs user action to spread.	Spreads automatically over network.
Effect	Corrupts files, slows system.	Consumes network resources.
Replication	Requires host.	Does not require host.
Example	Melissa, CIH	Conficker, Blar

7. What is a Trojan Horse?

A Trojan Horse is a type of malware that appears to be useful or legitimate software but performs malicious activities once installed.

8. Define Botnet.

A Botnet is a group of infected computers (bots) controlled remotely by an attacker ("botmaster") to perform malicious activities.

9. What is Cryptanalysis?

Cryptanalysis is the study and science of analyzing and breaking cryptographic systems without knowing the key.

Goal: To recover the plaintext or secret key ciphertext.

10. Define Cryptography.

Cryptography is the art and science of securing information by transforming it into unreadable form (ciphertext) using algorithms and keys.

11. Difference between Symmetric and Asymmetric encryption.

Feature	Symmetric (Secret Key)	Asymmetric (Public Key)
Keys	Same key for encryption and decryption.	Pair of keys: Public key for encryption, Private key for decryption.
Key Distribution	Difficult (secure channel needed).	Easy (public key can be shared).
Speed	Faster	Slower
Example	AES, DES	RSA, ECC

12. What is DES?

DES (Data Encryption Standard) is a symmetric block cipher that uses a 56-bit key to encrypt 64-bit data in blocks.

13. Why is DES considered weak?

- Uses short key length (56-bit).
- Brute force attack possible (2^{56} combinations).
- Advances in computing power can break it.
- Vulnerable to certain cryptanalytic attacks.

14. What is AES?

AES (Advanced Encryption Standard) is a symmetric block cipher that uses 128, 192 or 256-bit keys to encrypt 128-bit data in blocks. It is more secure and faster than DES.

15. List AES transformations.

- SubBytes (Non-linear substitution)
- ShiftRows (Row-wise shifting)
- MixColumns (Column mixing)
- AddRoundKey (XOR with round key)

16. Define Block Cipher and Stream Cipher.

Block Cipher: Encrypts fixed-size blocks of data (e.g., 64 or 128 bits) at a time.

Stream Cipher: Encrypts data one bit (or byte) at a time as a stream.

17. What is Key Expansion?

Key Expansion is the process of generating a set of round keys from the original cipher key. These round keys are used in each round of encryption.

18. Define RSA algorithm.

RSA is an asymmetric cryptographic algorithm based on number theory. It involves:

- Key Generation
- Encryption (using public key)
- Decryption (using private key)

Key Generation: Generate a public key (e, n) and a private key (d, n).

19. What is Public Key Cryptography?

Public Key Cryptography uses a pair of keys:

- Public Key:** Shared with everyone, used for encryption.
- Private Key:** Kept secret, used for decryption.

It solves the key distribution problem of symmetric encryption.

20. What is Cryptography?

Cryptography is the art and science of securing information by converting it into unreadable form to prevent unauthorized access and ensuring secure communication.

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